Theory of Common Waveform Estimation (CWE) Toolbox

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Assumption and Purpose

An EEG epoch of a channel, which is assumed to consist of waveforms common across trials and noise, is expressed by

$$y_n^{(ch)}(t) = \sum_{k=1}^{K} s_k^{(ch)}(t - \tau_{n,k}) + v_n^{(ch)}(t) \\ (t = 0, \cdots, T-1;\ n = 1, \cdots, N;\ ch = 1, \cdots, CH), \quad (1)$$

where $y_n^{(ch)}(t)$: observed EEG epoch of channel ch in trial n, $s_k^{(ch)}(t)$: k-th waveform of channel ch, $\tau_{n,k}$: delay of $s_k^{(ch)}(t)$ in trial n, $v_n^{(ch)}(t)$: noise of channel ch in trial n, and K: number of waveforms. Noise $v_n^{(ch)}(t)$ is assumed to be a stationary process.

For simplicity, we rewrite Eq. (1) as a matrix representation in the Fourier domain as below. By taking the discrete Fourier transform of Eq. (1), we obtain

$$Y_n^{(ch)}(\omega) = \sum_{k=1}^{K} \exp(-i2\pi\omega\tau_{n,k}/T) S_k^{(ch)}(\omega) + V_n^{(ch)}(\omega) \quad (\omega = 0, \cdots, T-1), \quad (2)$$

where $Y_n^{(ch)}(\omega)$, $S_k^{(ch)}(\omega)$, and $V_n^{(ch)}(\omega)$: the discrete Fourier transforms of $y_n^{(ch)}(t)$, $s_k^{(ch)}(t)$, and $v_n^{(ch)}(t)$, respectively. Eq. (2) is rewritten as

$$\boldsymbol{Y}^{(ch)}(\omega) = \boldsymbol{E}(\omega, \boldsymbol{\tau}) \boldsymbol{S}^{(ch)}(\omega) + \boldsymbol{V}^{(ch)}(\omega), \quad (3)$$

where

$$\boldsymbol{Y}^{(ch)}(\omega) = [Y_1^{(ch)}(\omega), \cdots, Y_N^{(ch)}(\omega)]^T,$$

$$\boldsymbol{E}(\omega, \boldsymbol{\tau}) = \begin{bmatrix} \exp(-i2\pi\omega\tau_{1,1}/T) & \cdots & \exp(-i2\pi\omega\tau_{1,K}/T) \\ \vdots & \ddots & \vdots \\ \exp(-i2\pi\omega\tau_{N,1}/T) & \cdots & \exp(-i2\pi\omega\tau_{N,K}/T) \end{bmatrix},$$

$$\boldsymbol{\tau} = \tau_{n,k}\ (n = 1, \cdots, N;\ k = 1, \cdots, K),$$

$$\boldsymbol{S}^{(ch)}(\omega) = [S_1^{(ch)}(\omega), \cdots, S_K^{(ch)}(\omega)]^T,$$

and

$$\boldsymbol{V}^{(ch)}(\omega) = [V_1^{(ch)}(\omega), \cdots, V_N^{(ch)}(\omega)]^T.$$

The purpose of the proposed method is to obtain $\hat{\boldsymbol{S}}^{(ch)}(\omega)$, $\hat{\boldsymbol{\tau}}$ and $\hat{K}$ only from $\boldsymbol{Y}^{(ch)}(\omega)$, where $\hat{\boldsymbol{S}}^{(ch)}(\omega)$: the estimated $\boldsymbol{S}^{(ch)}(\omega)$, $\hat{\boldsymbol{\tau}}$: the estimated $\boldsymbol{\tau}$, and $\hat{K}$: the estimated K.

Overview of Procedure for Obtaining $\hat{\boldsymbol{S}}^{(ch)}(\omega)$, $\hat{\boldsymbol{\tau}}$, and $\hat{K}$

The proposed method consists of three steps: *Delay estimation*, *Waveform estimation*, and *Evaluation*, which are consecutively repeated as shown in Fig. 1. At the beginning of the procedure, we set K to 1. In the *Delay estimation* step, we obtain $\hat{\boldsymbol{\tau}}$ using preset K. In the *Waveform estimation* step, we obtain $\hat{\boldsymbol{S}}^{(ch)}(\omega)$ using $\hat{\boldsymbol{\tau}}$ and preset K. In the *Evaluation* step, we evaluate whether preset K is true using $\hat{\boldsymbol{S}}^{(ch)}(\omega)$ and $\hat{\boldsymbol{\tau}}$. If preset K is evaluated as wrong, we return to the *Delay estimation* step by increasing preset K by 1. By repeating this procedure until preset K is evaluated as true, we can simultaneously obtain $\hat{\boldsymbol{S}}^{(ch)}(\omega)$, $\hat{\boldsymbol{\tau}}$, and $\hat{K}$.

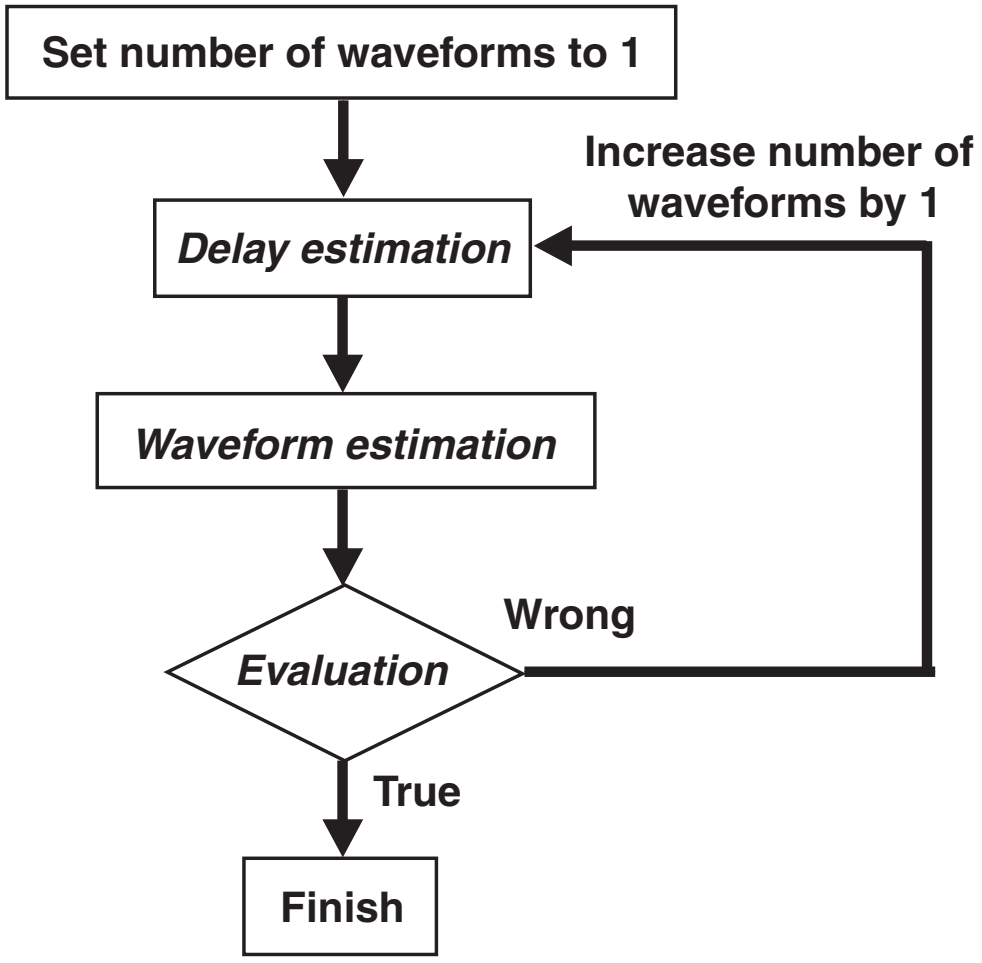


Figure 1: Flowchart to estimate number of waveforms common across trials, their delays in individual trials, and all of the waveforms.

The *Evaluation* procedure is based on the results of *Delay estimation* and *Waveform estimation*. The procedure of *Delay estimation* is based on that of *Waveform estimation*. Therefore, we describe these steps in the following order: *Waveform estimation*, *Delay estimation*, and *Evaluation*.

Waveform estimation

In this step, we estimate $\boldsymbol{S}^{(ch)}(\omega)$ from $\boldsymbol{Y}^{(ch)}(\omega)$ when $\boldsymbol{\tau}$ and K are given.

In the least squares method, estimated $\boldsymbol{S}^{(ch)}(\omega)$ is expressed by

$$\hat{\boldsymbol{S}}^{(ch)}(\omega) = \underset{\boldsymbol{S}^{(ch)}(\omega)}{\operatorname{argmin}} ||\boldsymbol{Y}^{(ch)}(\omega) - \boldsymbol{E}(\omega, \boldsymbol{\tau})\boldsymbol{S}^{(ch)}(\omega)||^2,$$

where $\underset{x}{\operatorname{argmin}} \cdots$ represents x that minimizes $\cdots$ and $||\cdots||$ represents the norm of $\cdots$. By solving

$$\frac{\partial ||\boldsymbol{Y}^{(ch)}(\omega) - \boldsymbol{E}(\omega, \boldsymbol{\tau})\boldsymbol{S}^{(ch)}(\omega)||^2}{\partial \boldsymbol{S}^{(ch)}(\omega)} = 0,$$

we obtain

$$\hat{\boldsymbol{S}}^{(ch)}(\omega) = [\boldsymbol{E}(\omega, \boldsymbol{\tau})^T \boldsymbol{E}(\omega, \boldsymbol{\tau})]^{-1} \boldsymbol{E}(\omega, \boldsymbol{\tau})^T \boldsymbol{Y}^{(ch)}(\omega). \tag{4}$$

When $\omega = 0$, all of the values in $\boldsymbol{E}(\omega, \boldsymbol{\tau})$ become 1, and $\boldsymbol{E}(\omega, \boldsymbol{\tau})^T \boldsymbol{E}(\omega, \boldsymbol{\tau})$ becomes singular. This corresponds to the fact that arbitrary constants can be added to $\hat{s}_k^{(ch)}(t)$. Therefore, we set the time average of $\hat{s}_k^{(ch)}(t)$ to be 0 by

$$\hat{\boldsymbol{S}}^{(ch)}(\omega) = \begin{cases} 0 & \omega = 0 \\ [\boldsymbol{E}(\omega, \boldsymbol{\tau})^T \boldsymbol{E}(\omega, \boldsymbol{\tau})]^{-1} \boldsymbol{E}(\omega, \boldsymbol{\tau})^T \boldsymbol{Y}^{(ch)}(\omega) & \omega \neq 0 \end{cases}. \tag{5}$$

We can obtain $\hat{s}_k^{(ch)}(t)$ by taking the inverse discrete Fourier transform of $\hat{\boldsymbol{S}}^{(ch)}(\omega)$.

Delay estimation

In this step, we estimate $\boldsymbol{\tau}$ from $\boldsymbol{Y}^{(ch)}(\omega)$ when K is given.

In the least squares method, estimated $\boldsymbol{\tau}$ is expressed by

$$\hat{\boldsymbol{\tau}} = \underset{\boldsymbol{\tau}}{\operatorname{argmin}} \sum_{ch=1}^{CH} \sum_{\omega=1}^{T/2} ||\boldsymbol{Y}^{(ch)}(\omega) - \boldsymbol{E}(\omega, \boldsymbol{\tau})\boldsymbol{S}^{(ch)}(\omega)||^2. \tag{6}$$

In this equation, since $\boldsymbol{S}^{(ch)}(\omega)$ is neither known nor determined by $\boldsymbol{\tau}$, we replace it by $\hat{\boldsymbol{S}}^{(ch)}(\omega)$ in Eq. (5), which is the least squares solution of $\boldsymbol{S}^{(ch)}(\omega)$ determined by $\boldsymbol{\tau}$. Then Eq. (6) is rewritten as

$$\hat{\boldsymbol{\tau}} = \underset{\boldsymbol{\tau}}{\operatorname{argmin}} \sum_{ch=1}^{CH} \sum_{\omega=1}^{T/2} ||\boldsymbol{Y}^{(ch)}(\omega) - \boldsymbol{E}(\omega, \boldsymbol{\tau})[\boldsymbol{E}(\omega, \boldsymbol{\tau})^T \boldsymbol{E}(\omega, \boldsymbol{\tau})]^{-1} \boldsymbol{E}(\omega, \boldsymbol{\tau})^T \boldsymbol{Y}^{(ch)}(\omega)||^2. \quad (7)$$

By solving Eq. (7), we can estimate $\boldsymbol{\tau}$. However, because $\boldsymbol{E}(\omega, \boldsymbol{\tau})[\boldsymbol{E}(\omega, \boldsymbol{\tau})^T \boldsymbol{E}(\omega, \boldsymbol{\tau})]^{-1} \boldsymbol{E}(\omega, \boldsymbol{\tau})^T$ in Eq. (7) is nonlinear with respect to $\boldsymbol{\tau}$, we cannot solve Eq. (7) analytically. We solve it by a hybrid optimization algorithm that consists of two consecutive stages: global search and local search. First, we obtain an approximate solution by global search and then obtain the optimal solution by local search. In global search, we conduct a random search (Zhigljavsky, 1991) (see Appendix) modified from that in our previous study (Takeda et al., 2008b) for M times with a different initial $\boldsymbol{\tau}$. Then we obtain M sets of $\boldsymbol{\tau}$ and $o_{\boldsymbol{\tau}}$ ($= \sum_{ch=1}^{CH} \sum_{\omega=1}^{T/2} ||\boldsymbol{Y}^{(ch)}(\omega) - \boldsymbol{E}(\omega, \boldsymbol{\tau})[\boldsymbol{E}(\omega, \boldsymbol{\tau})^T \boldsymbol{E}(\omega, \boldsymbol{\tau})]^{-1} \boldsymbol{E}(\omega, \boldsymbol{\tau})^T \boldsymbol{Y}^{(ch)}(\omega)||^2$) and select the $\boldsymbol{\tau}$ that minimizes $o_{\boldsymbol{\tau}}$. In local search, we conduct a grid search (see Appendix) by setting the $\boldsymbol{\tau}$ selected in global search as initial $\boldsymbol{\tau}$.

After optimization, we adjusted the averages across n of obtained $\boldsymbol{\tau}$. This adjustment is required because the averages of $\boldsymbol{\tau}$ vary depending on the time points defined as the onsets of the waveforms, and the onsets are arbitrarily determined in optimization. Adjusted $\boldsymbol{\tau}$ is referred to as $\hat{\boldsymbol{\tau}}$.

Evaluation

In this step, we evaluate whether preset K is true.

As described above, once K is set, we can obtain $\hat{\boldsymbol{\tau}}$ and $\hat{\boldsymbol{S}}^{(ch)}(\omega)$ by the steps of *Delay estimation* and *Waveform estimation*, respectively, and we can obtain the residual error between observed and reconstructed EEGs by

$$re_n^{(ch)}(t) = \mathrm{IDFT}\left[Y_n^{(ch)}(\omega) - \sum_{k=1}^{K} \exp(-i2\pi\omega\hat{\tau}_{n,k}/T) \hat{S}_k^{(ch)}(\omega) \right], \quad (8)$$

where IDFT$[\cdots]$: inverse discrete Fourier transform of $\cdots$. We evaluate preset K by examining $re_n^{(ch)}(t)$. When preset K is smaller than true, $s_k^{(ch)}(t)$ should remain in $re_n^{(ch)}(t)$. As a result,

$re_n^{(ch)}(t)$ should be a nonstationary process. In contrast, when preset K is true, $s_k^{(ch)}(t)$ should disappear from $re_n^{(ch)}(t)$. As a result, $re_n^{(ch)}(t)$ should be a stationary process. Therefore, we evaluate preset K by examining whether $re_n^{(ch)}(t)$ is a stationary process. We examine whether the distribution of $re_n^{(ch)}(t)$ differs before and after stimulus onsets. We divide all $re_n^{(ch)}(t)$ ($n = 1, \cdots, N;\ ch = 1, \cdots, CH$) into two samples corresponding to before and after stimulus onsets, and conduct a two-tailed Two-Sample Kolmogorov-Smirnov test to test the null hypothesis that the two samples are drawn from the same distribution. The probability of $p < 0.05$ is accepted as significant. If the null hypothesis is rejected, we regard preset K as wrong. If the null hypothesis comes to be accepted, we regard preset K as true.

Simulation Tests

To examine the performance of the proposed method, we conducted simulation tests for overall procedure.

In the simulation test, we generated simulated data $y_n^{(ch)}(t)$ as follows. We generated simulated data $y_n^{(ch)}(t)$ with the following parameters: number of waveforms $K = 3$, number of channels $CH = 1$, number of trials $N = 100$, and SNR= 6. The three waveforms, $s_1(t), \cdots, s_3(t)$, are generated by an exponential function, a cosine function, a rectangular function, respectively. All of the waveforms were identical regardless of channels ch. We set the variance across t of the waveforms to 1. We set the delays of $s_1(t)$ to 1 and generated the delays of the other waveforms by Gaussian random numbers [mean = 18, standard deviation (SD) = 5]. We used white noise as noise $v_n^{(ch)}(t)$. Then we generated $y_n^{(ch)}(t)$ from $s_k(t)$, $\boldsymbol{\tau}$ and $v_n^{(ch)}(t)$, based on Eq. (1). Figs. 2A–D show the original waveforms $s_k(t)$ ($k = 1, 2, 3$) and the noise $v_n^{(ch)}(t)$. Fig. 2E shows the simulated data $y_n^{(ch)}(t)$.

From the simulated data (Fig. 2E), we estimated K, $\boldsymbol{\tau}$, and $s_k(t)$ (Figs. 2F–H). Estimated K is correctly 3. Estimated $s_k(t)$ are highly correlated with the original ones; all of the correlation coefficients between the estimated and original $s_k(t)$ are higher than 0.99 (Figs. 2F–H). The RSMEs between the estimated and original $\boldsymbol{\tau}$ are 0, indicating that the estimation is completely accurate. These results indicate that the proposed method, as a whole, works well for the simulated data.

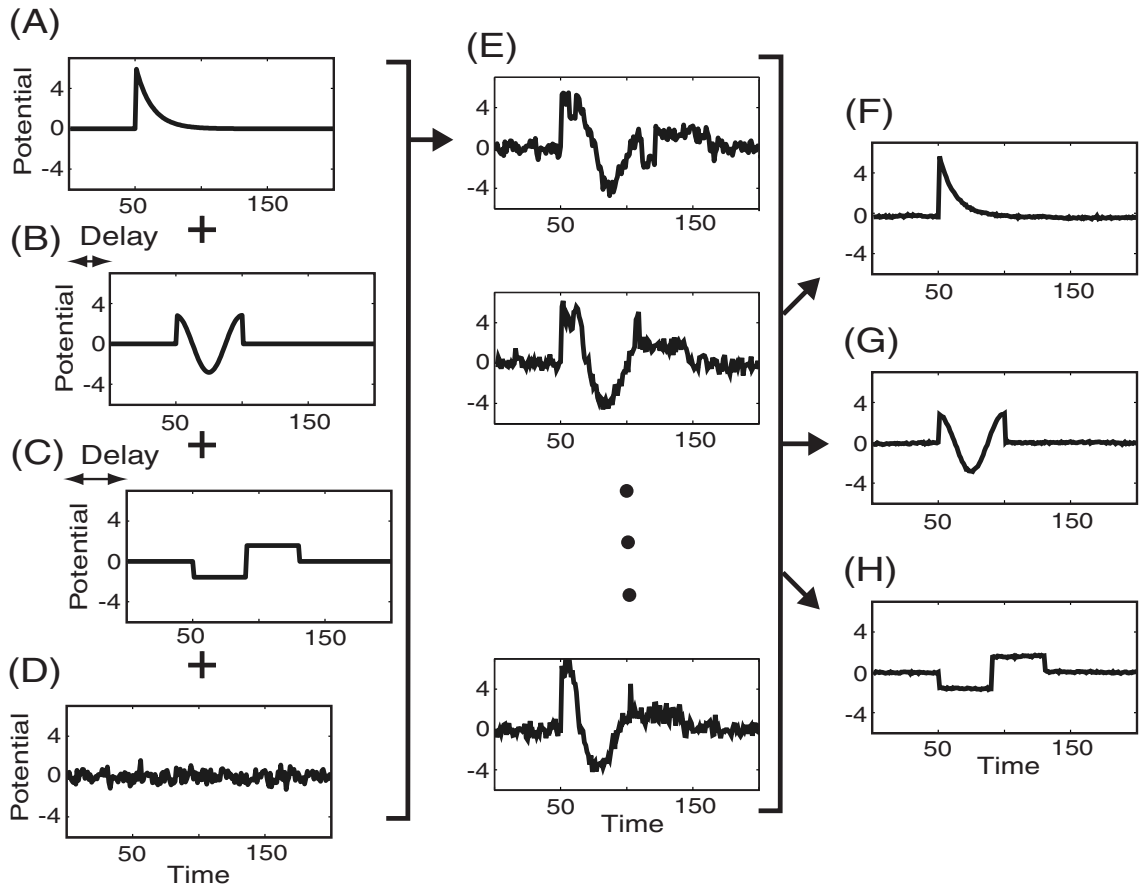


Figure 2: Simulation tests for overall procedure. (**A**–**D**): Original waveforms $s_k(t)$ and noise $v_n^{(ch)}(t)$. (**A**): $s_1(t)$. (**B**): $s_2(t)$. (**C**): $s_3(t)$. (**D**): $v_n^{(ch)}(t)$. (**E**): Simulated data $y_n^{(ch)}(t)$ generated from $s_1(t)$, $s_2(t)$, $s_3(t)$, and $v_n^{(ch)}(t)$. (**F**–**H**): Estimated waveforms $\hat{s}_k(t)$. (**F**): $\hat{s}_1(t)$. (**G**): $\hat{s}_2(t)$. (**H**): $\hat{s}_3(t)$.

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Appendix

The procedure for random search in global search is described in MATLAB style as follows:

Generate $\boldsymbol{\tau}$ by random numbers;
Obtain $o_{\boldsymbol{\tau}}$;
for iter = 1 : 10
  for k = 1 : K
    for n = 1 : N
      Make $\boldsymbol{\tau}'$ by changing $\tau_{n,k}$ in $\boldsymbol{\tau}$ randomly;
      Obtain $o_{\boldsymbol{\tau}'}$;
      if $o_{\boldsymbol{\tau}'} < o_{\boldsymbol{\tau}}$
        $o_{\boldsymbol{\tau}} = o_{\boldsymbol{\tau}'}$**;**
        $\boldsymbol{\tau} = \boldsymbol{\tau}'$**;**
      end
    end
  end
end.

The procedure for grid search in local search is described in MATLAB style as follows:

Set $\boldsymbol{\tau}$ selected in global search;
Obtain $o_{\boldsymbol{\tau}}$;
$o' = o_{\boldsymbol{\tau}} + 1$**;**
while $o_{\boldsymbol{\tau}} < o'$
  $o' = o_{\boldsymbol{\tau}}$**;**
  for k = 1 : K

for n = 1 : N

for t = τ_{min} **:** τ_{step} **:** τ_{max}

Make $\boldsymbol{\tau}'$ **by changing** $\tau_{n,k}$ **in** $\boldsymbol{\tau}$ **to t;**

Obtain $o_{\boldsymbol{\tau}'}$**;**

if $o_{\boldsymbol{\tau}'} < o_{\boldsymbol{\tau}}$

$o_{\boldsymbol{\tau}} = o_{\boldsymbol{\tau}'}$**;**

$\boldsymbol{\tau} = \boldsymbol{\tau}'$**;**

end

end

end

end,

end,

where τ_{min}: minimum value of $\tau_{n,k}$, τ_{step}: step of $\tau_{n,k}$, and τ_{max}: maximum value of $\tau_{n,k}$.

References

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